

Patrick H. Jacobs
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Richmond, VA
804.564.3992

EDUCATION

University of Virginia, School of Engineering and Applied Science

M.S. Mechanical Engineering, B.S. Mechanical Engineering, Material Science Minor

GPA: 3.74, Dean's List

Relevant Coursework: Nanomanufacturing, Fracture Mechanics, Machine Elements & Fatigue Design

Charlottesville, VA
August 2023 – Present

WORK EXPERIENCE

SOJO Industries, *Engineering Intern*

Summer 2026

- Designed and constructed a mechanical assembly from initial CAD layouts through physical wiring and system testing, while managing vendor costs and BOM inventory.
- Designed an infeed conveyor lane layout from scratch by creating both digital models and assembly instructions, improving machine output by more than 10%.
- Assumed a full mechanical engineer's responsibilities during site staff transitions, overhauling sheet metal assemblies under tight deadlines and creating new drawings for replacement parts.

Virginia Motorsports Formula SAE, *Mechanical Design Member*

August 2024 - Present

- Continuing development of CAD and CAM skills while building a basic understanding of vehicle dynamics and mechanical systems.
- Designed wheel uprights, starting from material choice through to shape optimization using finite element analysis and accounting for manufacturability.

BrownGreer PLC, *Claims Reviewer*

Summer 2024

- Worked on a team reviewing claims for the 3M Earplug lawsuit.

PROJECTS

Electric Bike Personal Project

July 2025 - Present

- Sourced electrical components and assembled an electric bicycle on a stock hardtail frame, with the goal of learning more about integrating electrical and mechanical systems.

UVA In Germany: *Global Technology Practices*

Summer 2025

- Studied abroad in Germany, visiting several manufacturing and headquarter sites of companies including Mercedes, Bosch, and BASF while learning about German culture and engineering.

LEADERSHIP, HONORS, AND EXTRACURRICULAR ACTIVITIES

First Robotics Competition Team 5804, *Team Member / Captain*

Spring 2019 –2023

- Collaborated with a team of 40+ people to construct a new robot each year to complete complex tasks in the annual First Robotics Competition.
- Competed and won several qualifier and district events. Qualified for the international competition and placed within the top 30 of 3,000+ teams in both the 2022 and 2023 seasons.
- Developed leadership and technical skills while dedicating 1,000+ hours of work over the course of 5 seasons.
- Developed mechanical and CAD modeling skills; learned basic FEA, CAM skills through usage of a CNC mill, and modeled and milled a majority of the team's custom parts. Received a Creativity Award at the international competition for the robot's design.

Leukemia and Lymphoma Society, *Student of the Year Candidate*

Spring 2021

- Led a team of 12 individuals to fundraise for children's blood cancer research. Raised roughly \$47,000 for blood cancer research.

GENERAL SKILLS

Technical: Solidworks CSWA, Autodesk Inventor, Fusion 360, Python, Matlab, Microsoft Excel, Microsoft Word

Design: Computer Aided Design and Manufacturing, 3 and 4 Axis machining, basic finite element analysis